

This protocol has been adapted from the Hughes lab protocol at Northeastern University for the Lotterhos lab eastern oyster seascape genomics project. We have made some substantial adjustments to sample volume, assay volumes, and have newly automated data analysis steps. We have also written this protocol with the aim of it being as clear and jargon-free as possible. Please reach out to MGE if there is any confusion about steps, and we will answer your question and adjust the protocol accordingly!

What is the goal of the multiplex disease qPCR assay?

Here, we are running eastern oyster DNA extractions through a molecular detection protocol for multiple parasites. This protocol allows us to detect which oysters were infected with each of two parasites, MSX and Dermo.

Glossary

MSX (aka *H.nelsoni*, *Hnel*): An oyster microparasite that we are detecting in the assay
Dermo (aka *P.marinus*, *Pmar*): An oyster microparasite that we are detecting in the assay

Gblocks: ultra-concentrated parasite DNA that we use to make a standard dilution

Positives: concentrated parasite DNA that was extracted from environmentally-infected oysters

Negatives (aka *NTC*): Non-template control, or true negative, which is just sterile water!

Probes: light-sensitive fluorescent probe that attaches to amplified DNA and makes it glow, which the machine quantifies. Probes help us tell us how much DNA is in the well, which is the result that we're interested in.

Primers: DNA templates that bind to and help to amplify DNA. Primers come in Forward (F) and Reverse (R) configurations. Primers are species-specific, so we have a unique set for each of MSX and Dermo.

20x Assay: This is the mix of primers, probes, and water for each disease. We make this in advance, and it gets added to the MasterMix to make the Working Mix.

MasterMix: pre-made duplex master mix, it arrives ready-to-go and we just make aliquots

Working Mix: the combination of MasterMix, 20x Assay primer/probe mix, and water that goes into every single well on the plate. This is the stuff that makes the assay run!

Serial dilution: Progressively diluting a sample across a set of tubes to make a lower and lower concentration of DNA in each.

Multichannel: Pipette with 8 tips

Ordering materials & what to do when materials arrive

MasterMix

When we order the 5 mL volume, we can run 8 plates with this amount. We make 8 aliquots of 625 ul each into 1.5 mL low-bind tubes and immediately store them in the 5C fridge. Each plate requires 618 ul of MM, so there is a small amount of buffer room for pipetting error. Do not

contaminate the mastermix, it is the most expensive material for the plate! It should be pipetted in a separate sterile workspace away from any other qPCR stuff.

Primers and probes (making combined 20x Assay)

Probes are light-sensitive and degrade when exposed to light. To mitigate this, we immediately wrap probes in aluminum foil to block light when they arrive. Keep them wrapped at all times when you are not actively pipetting. Probes are also sensitive to freeze-thaw cycles, so it's important to think in advance about many plates you need to run in order to minimize the freeze-thaw cycles and light exposure of your probes. The primer/probe mixes for each disease should be prepared separately.

To make a small batch of 20x Assay (enough to run ~1 plate or a test run) of primer/probe mix, combine the following materials for each disease into their own Qubit-sized (tiny!) tube.

Single Dermo Mix (1 plate) 18 ul Dermo (PmarFav) F primer 18 ul Dermo (PmarFav) R primer 5 ul Dermo (PmarFav) probe 59 ul 1x TE Buffer Single MSX Mix (1 plate) 18 ul MSX (Hnel) F primer 18 ul MSX (Hnel) R primer 5 ul MSX (Hnel) probe 59 ul 1x TE Buffer
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However, when I receive materials, I like to **scale up the recipe to make as much material at one time as possible**. This is particularly efficient if you are wanting to run multiple plates. The nice thing about preparing the scaled mix in advance is that you can prepare vials with the exact amount needed for 1 plate of each primer/probe mix, so now you will only need to thaw 1 vial of MSX primer/probe mix and 1 vial of Dermo primer/probe mix for each plate that you run.

Here is an example of a recipe scaled up to run 14 total plates (so all values from above are multiplied x14). To make this scaled batch, combine the following materials for each disease into their own 1.5 mL tube - do NOT use a tiny Qubit tube, the volume is too large.

Scaled Dermo 20x Assay Mix (14 plates) 252 ul Dermo (PmarFav) F primer 252 ul Dermo (PmarFav) R primer 70 ul Dermo (PmarFav) probe 826 ul 1x TE Buffer Scaled MSX 20x Assay Mix (14 plates) 252 ul MSX (Hnel) F primer 252 ul MSX (Hnel) R primer

70 ul MSX (Hnel) probe 826 ul 1x TE Buffer

Once you've made your entire scaled recipe, **we pipette aliquots of 62 ul of each 20x Assay into Qubit-sized (tiny!) tubes** labeled with either 'MSX' or 'D' for Dermo. These tubes should be immediately wrapped in foil and placed in a freezer box, where they will be held at -20. These aliquots are the final 20x Assay that will go into making the working mix for each plate.

Positives

We have in the past received Dermo and MSX positives from the Carnegie Shellfish Pathology lab at the Virginia Institute of Marine Science. The best way to get in contact with the lab is to email the lab manager (as of 2025, this is Melissa Butler mcbutler@vims.edu, and cc Ryan Carnegie in the email). We request the positives and they provide them free of charge, but we have to pay for and send them a pre-paid FedEx shipping label. In an email exchange, we were told "We have been storing them in the fridge and they should be stable at room temperature for a bit as they are dried DNA pellets, though you may want to consider express shipping so that they are not sitting in a hot warehouse for an extended period."

Upon arrival, the positives were tiny dried pellets of DNA in 1.5 mL tubes. To rehydrate:

- Add 25 ul of 1x TE to each vial to rehydrate the pellet
- Heated the vials to ~50 C to help the pellet dissolve
- Occasionally vortex and spin down during the heating process
- This will probably take 20-30 minutes, but visually re-assess to check for dissolution

Dermo is more concentrated in the environmentally-infected true positives because the intensity of infection is generally much higher than with MSX in oysters. Because of this, we dilute the Dermo positive 1:100, while we dilute the MSX 1:10.

We don't know the exact concentration of the true positives, and this is fine. The point is simply to dilute them by a rough estimate so that we don't have more parasite DNA than what we need in each plate. Dermo is more concentrated, so we dilute it further. These dilutions worked very well for us in prior plate runs, we never had a positive well read as negative.

To dilute:

- Add 10 ul of the rehydrated MSX positive to 90 ul of molecular-grade water
- Add 1 ul of the rehydrated Dermo positive to 99 ul of molecular grade-water

These concentrations are now the final positives that can go directly into each qPCR plate. I never had issues with freeze/thaw cycles, but you could make aliquots if preferred.

Gblocks (Gene Blocks) & Standards

Work with concentrated gBlock standards only in the hood.

Gblocks contain ultra-concentrated parasite DNA. This is troublesome because they can VERY easily aspirate (aspiration = tiny parasite particles moving through the air) and “infect” other tubes and surfaces. We don’t want this to happen, because it will lead to false positive results.

To ensure there is no contamination, only open these tubes in a hood, ideally in another lab away from where the plates are typically run. Use a box of tips dedicated specifically to parasite disease analysis!!! Do NOT have other tubes out while you have the GBlocks open.

When we receive the Gblocks, the new stock in the tube is dry. We generally follow the [the instructions provided by IDT](#) to make 10 ng/μl solution for each MSX and Dermo, with the following adaptations:

- Spin the dry tube down for ~10-15 seconds in a centrifuge at any speed
- Add 25 μl 1X TE Buffer (pH 8.0) to gBlock standard (250 ng/tube) to make a 10 ng/μl solution (*do not resuspend in water!*)
- Vortex briefly
- Incubate at 50C for 20 minutes (*do not skip this step!*)

Now you have your resuspended 10 ng/ul solution. It is in 25 ul of 1x TE buffer. However, we need to dilute further to get closer to the concentration needed for the start of our series.

- Add 225 ul 1x TE to bring the total volume up to 250 ul. **This makes a 1 ng/μl solution.**
- Repeat these steps for either the Dermo GBlock standard or the MSX GBlock standard, whichever you haven’t done yet, so that you have 1 ng/μl solutions of each.

At this point, if you won’t be running a qPCR plate in the near future, you can freeze your 1ng/ul solutions in the -20 freezer. If you are running plates soon, it makes sense to now make the standard dilution series by **combining** MSX and Dermo 1ng/ul solutions with 1x TE to make the **0.1 ng/ul standard**. The reason that we combine the two is because our standard series will only occupy one column of our qPCR plate, so we need to add the Dermo and MSX together.

Combined 0.1 ng/ul standard 10 μl of 1 ng/μl Dermo gBlock standard 10 μl of 1 ng/μl MSX gBlock standard 80 μl of 1X TE Buffer
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The 0.1 ng/ul standard is the top of our dilution series that goes on the plate (e.g., well A1 in the  qPCR_runs in the Unk-plate-layout tab).

The rest of the standard dilution series can be made by doing a serial dilution, where we progressively make a smaller concentration across 6 additional tubes (7 total, including your current 0.1 ng/ul standard). Label your 0.1 ng/ul standard tube with “Standard 1”. Set up 6 more 1.5 ul low-bind tubes and label them with “Standard 2”, “Standard 3”, etc. until “Standard 7”.

- In the remaining 6 tubes (Standard 2-7), add 90 ul 1x TE buffer to each.
- Pipette 10 ul from the Standard 1 (0.1 ng/ul) and add into the Standard 2 tube.
- Vortex the Standard 2 tube for 10-15 seconds
- Pipette 10 ul from the Standard 2 tube and add into the Standard 3 tube.
- Vortex the Standard 3 tube for 10-15 seconds.
- Pipette 10 ul from the Standard 3 tube into the Standard 4 tube.
- Vortex the Standard 4 tube for 10-15 seconds
- Repeat until you are finished and finish the Standard 7 tube

Now you should have 7 tubes with qPCR standards ranging from 0.1 ng/μl to 0.0001 pg/μl. It is possible to stop here, and simply freeze your bulk standard dilutions at -20. **These are the final concentrations that you will need to run the qPCR plate.**

However, freeze-thaw cycles can cause standard concentrations to change over time. To avoid this issue, ***I highly recommend making aliquots of the standards*** into individual 1.5 ml low-bind tubes. Each time we run a plate, we need 2.33 ul of each standard. I tend to aliquot anywhere from 3-5x this amount (so a total of 7 - 15 ul standard, plus a little extra for evaporation, pipetting error, etc.) into each aliquot tube. I have found that any more than 5 freeze-thaw cycles is likely to cause issues with your standards.

Here is the calculation of copy number from gBlock concentration for each parasite.

Willbur MSX			fmoles/ng =	12.97
Concentration (ng/μl)	Concentration (pg/μl)	Copy Number	Copy Number	
10	10,000	7.81E+10	78105340000	
1	1000	7.81E+09	7810534000	
0.1	100	7.81E+08	781053400	
0.01	10	7.81E+07	78105340	
0.001	1	7.81E+06	7810534	
0.0001	0.1	7.81E+05	781053	
0.00001	0.01	7.81E+04	78105	
0.000001	0.001	7.81E+03	7811	
0.0000001	0.0001	7.81E+02	781	
DeFav Dermo			fmoles/ng =	11.58
Concentration (ng/μl)	Concentration (pg/μl)	Copy Number	Copy Number	
10	10,000	6.97E+10	69734760000	
1	1000	6.97E+09	6973476000	
0.1	100	6.97E+08	697347600	
0.01	10	6.97E+07	69734760	
0.001	1	6.97E+06	6973476	
0.0001	0.1	6.97E+05	697348	
0.00001	0.01	6.97E+04	69735	
0.000001	0.001	6.97E+03	6973	
0.0000001	0.0001	6.97E+02	697	

1X TE: If we received a bottle of 20X TE to dilute, make a 1X TE concentration in a 50 mL aliquot tube by adding 2 mL of 20X TE into 38 mL of sterile molecular-grade water. This solution is stable long-term, so it can live on the aliquots shelf. Make sure to date and add your initials to the tube. Then this aliquot tube can live in the hood where you are running the qPCR plates in a tube rack.

Setting up a plate run (25 minutes)

1. Turn on the UV light in the Gatins lab hood and allow it to run for at least 20 minutes. The UV should have also been run at the completion of the last plate, but if not, allow it to run for the full hour cycle.

- a. Double-check that the single channel and multichannel repeater pipettes you will be using are inside of the hood while the UV light treatment is on.
2. Close the window shade next to the hood. Dim the lights in the lab by turning on only the lights above the sink or the set in the front of the lab by the door.
3. Navigate to the qPCR_runs datasheet ( qPCR_runs) and plan out which samples you are going to put on the plate today. Duplicate our plate setup, add the date, your initials, and then change the sample names accordingly. Do not move the positives, negatives, or standards around on the plate.
4. Gather the following materials and place them outside of the hood: roll of aluminum foil, counter-balance empty qPCR plate, sterile qPCR plate, sterile qPCR plate tape sheet, empty pipette tip waste container, lab notebook, sharpies
5. Gather the following materials and place them inside of the hood: sterile molecular-grade water, green plastic block PCR strip tube holder, and 6 strips of USA scientific PCR tubes with the attached cap (located on Gatins lab shelf above the desktop computer)
6. Set the multichannel repeater pipette to an uptake speed of 3, expel speed of 4, and volume of 10 ul with 2x steps.
7. Set the single channel repeater pipette to an uptake speed of 4, expel speed of 4, and volume of 21 ul with 3x steps.
8. Grab ice in two buckets, one small enough to fit in the hood, and a larger one on the counter beside the hood.
9. Remove the 625 ul aliquot tube of MM from the 5C Lotterhos lab fridge and place it on ice outside of the hood.
10. Remove plate(s) of DNA that you are working with from the -20 C Lotterhos lab freezer and place it on ice outside of the hood.
11. Remove the standards, primers/probe assay mixes, positive controls, and put on ice inside of the hood.
 - a. Only remove one set of the standards from the freezer at a time. They are in 12 ul aliquots, so freeze/thaw the same set until it is finished before moving on.
 - b. The assay mixes should be in foil, do not unwrap them yet.

Pipetting the plate (45 minutes)

Once you start this process, there is not a good stopping point. Make sure that you are prepared to execute this entire series of pipetting steps before you start. The assays are light-sensitive.

1. Make the working mix (**WM**) in a sterile low-bind 1.5 mL tube.

618 ul TaqMan MasterMix
370.8 ul Molecular Grade H2O
61.8 ul 20x Dermo Assay primer/probe mix
61.8 ul 20x MSX Assay primer/probe mix

2. Vortex WM. Immediately wrap the lower portion of the tube in foil to prevent light exposure.
3. Using the 5-100 ul repeater single-channel pipette, add **21 ul** of WM into each tube.

- a. Ensure steps are set to 3x. Intake speed is 4, dispensing speed is 4.
 - b. Tip: With the electronic pipettes, the first “dispense” click will always “waste” 1uL. After aspirating the WM into the tip, stay in the tube to “waste” that first droplet. Then complete 3 pipettes into strip tube wells. To empty any leftover overage, click the “down” button 3x, this will dispense any leftover WM back into the WM tube.
 - c. Repeat, if it starts to get bubbly at any point, grab a new tip.
4. At the completion of each strip of tubes, move the entire strip under a piece of foil to prevent light exposure.
 - a. Keep the foil covering all tubes and keep all strip tube caps closed unless you are actively pipetting into them from this point on.
5. Add **2.33 ul** molecular grade water (negative control) into A6
6. Warm the first two standards in your hand, vortex for 5 sec, then briefly spin down.
7. Pipette **2.33 ul** standard one (0.1) into A1 and **2.33 ul** standard two (0.01) into B1. Generally, we pipette this in descending order, starting at the highest concentration and going to the lowest concentration. However, if you ever see weird results in your standard curve, you can try pipetting from lowest concentration and go to the highest concentration to avoid potential contamination at lower concentrations.
 - a. Return the standard aliquots to ice as soon as you are finished pipetting them.
8. Repeat with standards in sets of two. Remember that standard 7 is pipetted twice total, once each into G1 and H1.
9. Pipette **2.33 ul** DNA sample into each well according to the plate plan. Vortex each sample before pipetting by placing your hand on the vortexer and holding the sample out. If pipetting a full column of samples, remove and replace the rubber caps completely.
 - a. Use a counter-weight of another CviMVP frozen plate (-20 C Lotterhos freezer) to spin down using the centrifuge in the common lab space
10. Pipette **2.33 ul** Dermo positive into G6
11. Pipette **2.33 ul** MSX positive into H6
12. Vortex each strip tube by placing your hand on the vortexer, then spin down.
13. Place the clear plastic white-skirted qPCR plate on ice inside the hood.
14. Using the multichannel repeater pipette, transfer 10 ul into each well of the plate.
 - a. Ensure steps are set to 2x. Intake speed is 3, dispensing speed is 4.
 - b. **Tip:** Place the pipette tips at the very bottom of the well. Make sure they are completely even and all pressed against the plastic tubes. After you pipette up, expel the first ‘waste’ pipette back into the strip tubes. All volume should now look even in the pipette tips. Pipette down once into the first column, then down once into the second column. There should be a tiny amount of fluid left in the tips, this is ok. Check for consistency across all wells by holding the plate up.
15. Keep foil covering the columns once you have finished pipetting.
16. Cover the entire plate with a clear plastic tape sheet
17. Vortex gently for 5-10 sec and spin down for 5-10 sec using an empty counterweight

0.1
0.01
$1e^{-3}$
$1e^{-4}$
$1e^{-5}$
$1e^{-6}$
$1e^{-7}$
$1e^{-7}$

Baseline Plate Layout

0.1	0.1									NTC	NTC
0.01	0.01										
1e ⁻³	1e ⁻³										
1e ⁻⁴	1e ⁻⁴										
1e ⁻⁵	1e ⁻⁵										
1e ⁻⁶	1e ⁻⁶										
1e ⁻⁷	1e ⁻⁷									Dermo +	Dermo +
1e ⁻⁷	1e ⁻⁷									MSX +	MSX +

Running the qPCR plate (45-60 minutes)

We are using the Hughes lab qPCR machine while we evaluate potential contamination in the common lab space Lotterhos qPCR machine.

1. Turn on the machine with the power button on the back right-hand side. It takes a few minutes to warm up.
2. Open the lid by pressing the button in the center of the machine and load the plate, keeping the plastic sheet on the wells. Double-check the plate orientation (A1 in top left hand corner). Close the lid.
3. Select the 'Oyster' protocol and make sure that the run date correctly displays the date.
 - a. IMPORTANT: check the "all channels" box when prompted. This is NOT the default setting, and failure to check this box means the samples will not be read.
4. Press run. There is a warmup period, and then the thermocycler will begin to count down the amount of time remaining. It's a good idea to watch the machine for a minute or two to make sure it advances out of the warmup period.
5. The run takes about 45-60 minutes to complete.

Evaluating run quality (25 minutes)

1. When the run is finished, plug in the Hughes lab USB stick into the right hand side of the machine. It will flash red, wait until it glows solid red.
2. Press 'Export' run. Wait for the USB stick to flash red and return to a solid red glow.
3. Plug the USB stick into the Hughes lab laptop (no password necessary) and copy the Bio-Rad **compressed** run file from the USB folder to a new folder (create one using the date of the run and your initials) in 'Eppley_qPCR_oyster' Desktop folder.
4. Right click on the file and select 'Open' - this will open it in the BioRad CFX Maestro software program. Then you will be prompted to then save the **prcd** run file into the same 'Eppley_qPCR_oyster' Desktop folder. This is correct, click save. The BioRad CFX Maestro software program should now auto-open with your run file.

- a. Note: If you simply open the software first and then choose 'Browse', it will NOT work to open the file. I think this is due to the compressed file format.
5. Upload the Plate Setup Plan
 - a. Click the Plate Setup menu in the upper right-hand corner of the Data Analysis Page. Click "Replace Plate File" and apply the Bio-Rad Plate File called plate_layout_file_MGE that is stored in the Eppley_qPCR_oyster folder on the Desktop (and also stored in the Eppley_qPCR folder on the USB stick).
 - b. In this file, Replicates = 2, Unknown sample #s are 1-37 and standard dilutions are a series of 7 dilutions with the 7th repeated twice.
 - c. If running the 7th standard only once, open the same plate file, select the 2 wells at the bottom with the 8th standard and press "Clear Wells" button. Then select "Unknown", check the FAM and HEX boxes, and go to dilution series. In the dilution series, make it a replicate of 2, and start the numbering at #38. This will make that well position Unknown #38.
6. Eliminate the top standard and one of the bottom standards, whichever pair is least consistent with each other. The plate file has it auto-set to eliminate standard 8, but evaluate if standard 7 or standard 8 is higher quality.
7. Eliminate one other standard value based on deviation from the regression line or deviation between the points.
8. If any samples amplified early and not any further, this is likely machine error. Drag the horizontal bar on the log scale transformation above these curves. It may improve the efficiency value slightly.
9. Evaluate R^2 value and E value
 - a. Generally, we want the R^2 value to be at 0.99 or above. I try to adjust different parameters, like moving the threshold, or eliminating a different standard in order to achieve this.
10. Double-check that the negatives are negative, and the positives amplified. If the negatives did amplify, check to ensure that the MSX positive did not amplify for Dermo, and vice-versa. If that is true, the plate can still be used.
11. If all thresholds were passed, eliminate any samples from the plate that look like machine error or are vastly inconsistent (e.g. one rep amplified and the other did not).

Data analysis preparation (10 minutes)

Once the following sequence of steps is done you can be done for the day, the data analysis can be completed at another time. While the Hughes lab qPCR machine does retain run data, these steps ensure that a copy of all data we need to analyze is saved in our google drive.

1. Export the final data file using the 'Export' option and choose Excel 2007. Select the 'Eppley_qPCR' folder on the computer and press export. It will take a second to load, but a large set of files should now appear.
2. The only file that is necessary for the downstream R script is located in the 'Quantification Cq Results' output file from the machine run. Locate this file and save it on the USB stick.

3. Double check that all final files are saved to the USB stick and the Hughes Lab laptop in the Eppley_qPCR and Eppley_qPCR_oyster folders. Don't delete any of these files - keep them backed up in both places.
4. Using the Gatins lab laptop (password: 58877), open google drive and create a new folder with the plate number and date of the run.
5. Plug in the USB stick into the Gatins lab laptop and upload the Quantification Cq Results file to the new google folder.
6. Transfer other run metrics, like the .xlsx files and the .pcrd run file to the Google Drive. Right now, we don't have a use or plan for these, but we might need them down the line for other types of data analysis or for publication.

Data analysis

Two R scripts are available to process the raw Cq file outputs from the BioRad machine. The first is named 01_combine_qpcr.R and the second is named 02_filter_qpcr.R. Scripts are available in the GitHub/ModelValidationProgram/Oyster_Seascape/disease folder. To run the scripts, read in your Cq output file(s) and adjust the number of plates that go into the for loop in the 01_combine_qpcr.R script. This script will produce a few .csv files, so set your working directory accordingly. This script will combine all of your data and produce some raw aggregate files.

Once the 01_combine_qpcr.R script runs, execute the 02_filter_qpcr.R by setting the file pathway to wherever you stored the output .csv files. Then, run the script, which filters out any duplicate samples that you may have run, low-quality results, and runs further analysis by groups or populations that you may be interested in seeing.

Final output files will produce reports on individual samples, sample groups or populations, and overall on the entire dataset.